

GUARDRAIL LENGTH-OF-NEED CALCULATIONS

Bridge approach guardrail based on MDOT Roadway Design Manual Table 9-6-A and GR-4 dimensions.

Project	108388/101000	Job Number	SP-0016-02(040)
Route / Location	State Route 8	Calculation Date	2026-06-22
Prepared By	CW	Checked By	CW

DESIGN INPUTS

Facility	Two-Lane, Two-Way Roadway	Design Speed	65 mph
Design ADT	4200	Runout Length, LR	250.0000 ft
Hazard Offset, LA	30.0000 ft	Near-Side Offset, L2	8.0000 ft
Opposing-Side Offset, L2	20.0000 ft	Lane Width	12.0000 ft
Transition Length, L1	18.1458 (18'-1.7500")	Terminal Section	25.0000 ft
Flare Rate, a/b	30/1	Guardrail Increment	12.5000 ft

FINAL DESIGN OUTPUTS

A - Near Side	B - Near Rail	C - Opposing Side	D - Opposing Rail
155.6458 ft	112.5000 ft	80.6458 ft	37.5000 ft
Near Side Including Gating	168.1458 ft	Opposing Side Including Gating	93.1458 ft

SUPPORTING RESULTS

Near Minimum X	147.4230 ft	Near Required Rail, B	104.2772 ft
Opposing Minimum X	69.1621 ft	Opposing Required Rail, D	26.0163 ft
Rounded Rail, B	112.5000 ft	Rounded Rail, D	37.5000 ft

Design basis: MDOT Roadway Design Manual, Table 9-6-A, Barrier Length of Need. Calculated guardrail lengths are rounded in accordance with the existing 12.5000-foot increment workflow.

Engineering disclaimer: This calculation package documents user-entered inputs and computed results. It is not a sealed or stamped engineering document and must be independently reviewed for project applicability.

DETAILED CALCULATIONS

Project: 108388/101000 | Job Number: SP-0016-02(040)

COMMON PARAMETERS

Design Speed	65 mph	Design ADT	4200
LA	30.0000 ft	L1	18.1458 (18'-1.7500")
Terminal Section	25.0000 ft	LR	250.0000 ft

NEAR SIDE - OUTPUTS A AND B

Near-Side Length of Need

Minimum X	147.4230 ft	Required Rail Length	104.2772 ft
Rounded Rail Length	112.5000 ft	Final Overall Length	155.6458 ft

Length-of-need equation and substitution

$$X = (30.0000 + 0.0333 \times 18.1458 - 8.0000) / (0.0333 + (30.0000/250.0000))$$

B Required	$B(\text{required}) = 147.4230 - 18.1458 - 25.0000 = 104.2772$	104.2772 ft
Output B	$B = \text{round up max}(12.5000, 104.2772) \text{ to } 12.5000\text{-ft increment} = 112.5000$	112.5000 ft
Output A	$A = 112.5000 + 18.1458 + 25.0000 = 155.6458$	155.6458 ft

OPPOSING SIDE - OUTPUTS C AND D

Opposing-Side Length of Need

Minimum X	69.1621 ft	Required Rail Length	26.0163 ft
Rounded Rail Length	37.5000 ft	Final Overall Length	80.6458 ft

Length-of-need equation and substitution

$$X = (30.0000 + 0.0333 \times 18.1458 - 20.0000) / (0.0333 + (30.0000/250.0000))$$

D Required	$D(\text{required}) = 69.1621 - 18.1458 - 25.0000 = 26.0163$	26.0163 ft
Output D	$D = \text{round up max}(12.5000, 26.0163) \text{ to } 12.5000\text{-ft increment} = 37.5000$	37.5000 ft
Output C	$C = 37.5000 + 18.1458 + 25.0000 = 80.6458$	80.6458 ft

All displayed calculated lengths are shown to four decimal places. Internal calculation precision and the established 12.5000-foot increment rounding method are unchanged.

APPENDIX A

MDOT Roadway Design Manual Reference

Embedded reference document included for calculation traceability.

**Table 9-6-A
BARRIER LENGTH OF NEED**

Design Speed (mph)	Design Traffic Volume (ADT)				Shy Line Offset (ft)	Steepest Flare Rate for Barrier Inside Shy Line	Steepest Flare Rate for Barrier Beyond Shy Line	
	Over 10,000	5000-10,000	1000-5000	Under 1000			Rigid (Concrete Barrier)	Semi-Rigid or Flexible (W Beam, Thrie Beam)
	Runout Length L _R (ft)	Runout Length L _R (ft)	Runout Length L _R (ft)	Runout Length L _R (ft)				
70	360	330	290	250	9.0	30:1	20:1	15:1
65	330	290	250	225	8.5	28:1	19:1	15:1
60	300	250	210	200	8.0	26:1	18:1	14:1
55	265	220	185	175	7.0	24:1	16:1	12:1
50	230	190	160	150	6.5	21:1	14:1	11:1
45	195	160	135	125	6.0	18:1	12:1	10:1
40	160	130	110	100	5.0	16:1	10:1	8:1
35	135	110	95	85	4.5	15:1	9:1	8:1
30	110	90	80	70	4.0	13:1	8:1	7:1

Equations to determine end of barrier need:

Flared Design

$$X = \frac{L_A + \frac{b}{a}(L_1) - L_2}{\frac{b}{a} + \frac{L_A}{L_R}}$$

$$Y = L_A - \frac{L_A}{L_R}(X)$$

Non-flared Design

$$X = \frac{L_R (L_A - L_2)}{L_A}$$

$$Y = L_2$$

Definition of Terms:

X = length of need in advance of the area of concern

Y = lateral offset from edge of traveled way

b/a = barrier flare (e.g., 1/30)

L_c = recommended clear zone (feet)

L_A = distance from edge of traveled way to back of hazard (feet)

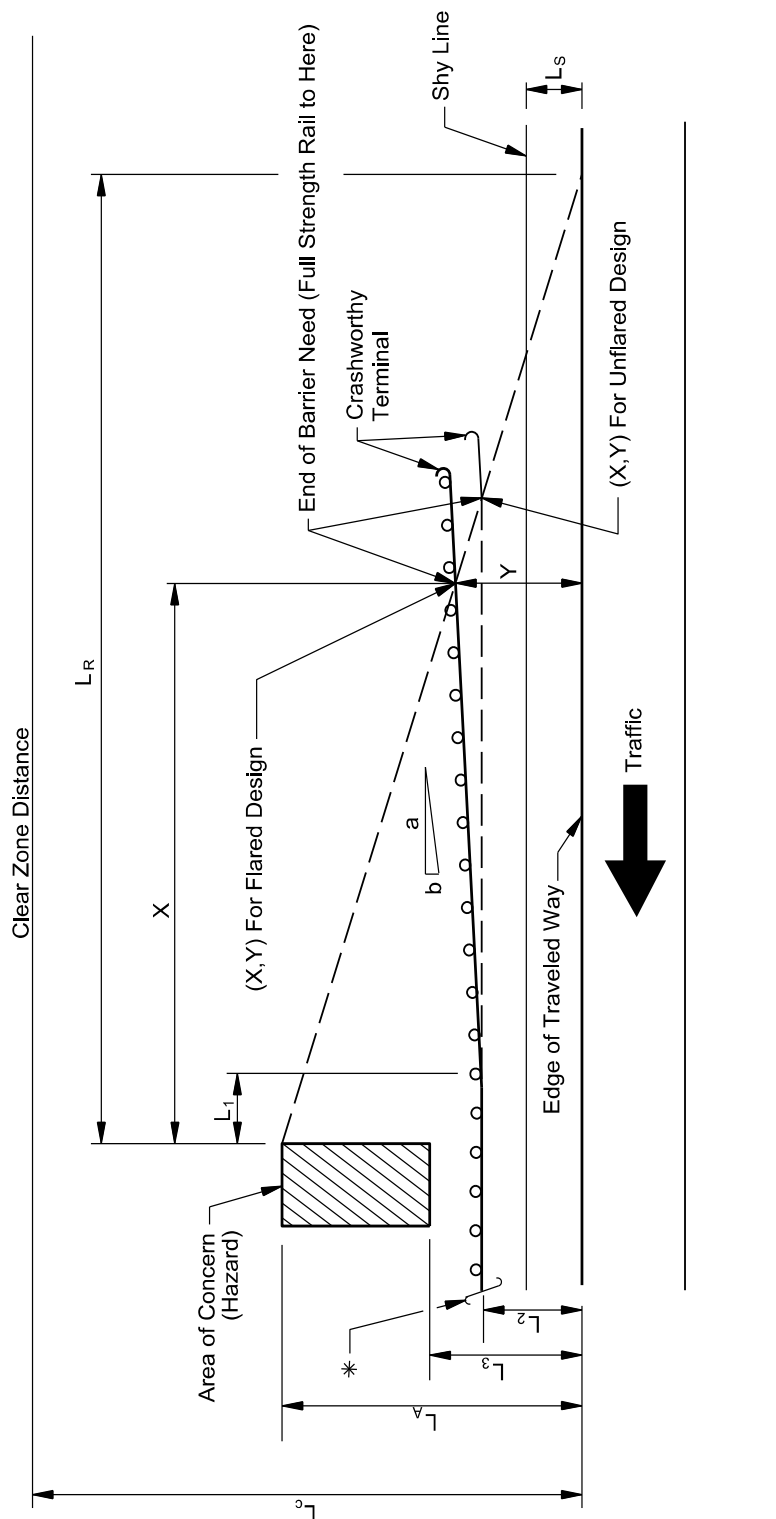
L_s = shy line offset, or distance at which barrier is no longer perceived as a hazard by a driver (feet)

L_R = runout length (feet)

L₁ = distance from hazard to where barrier flare begins (feet)

L₂ = distance from edge of traveled way to barrier (feet)

L₃ = distance from edge of traveled way to front of hazard (feet). (L₃ - L₂) should equal or exceed deflection distance.



BARRIER LENGTH OF NEED
Figure 9-6-A

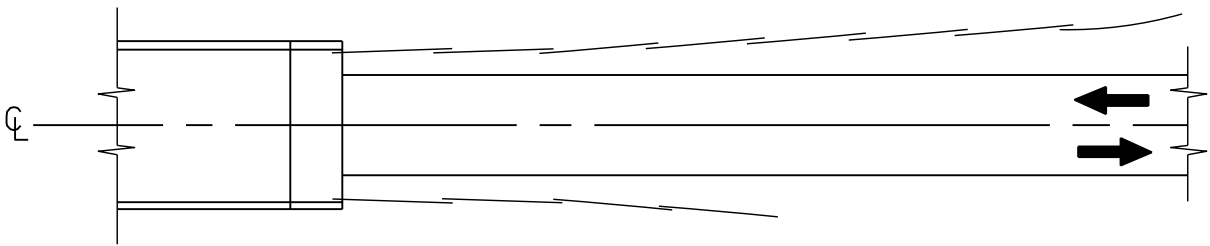
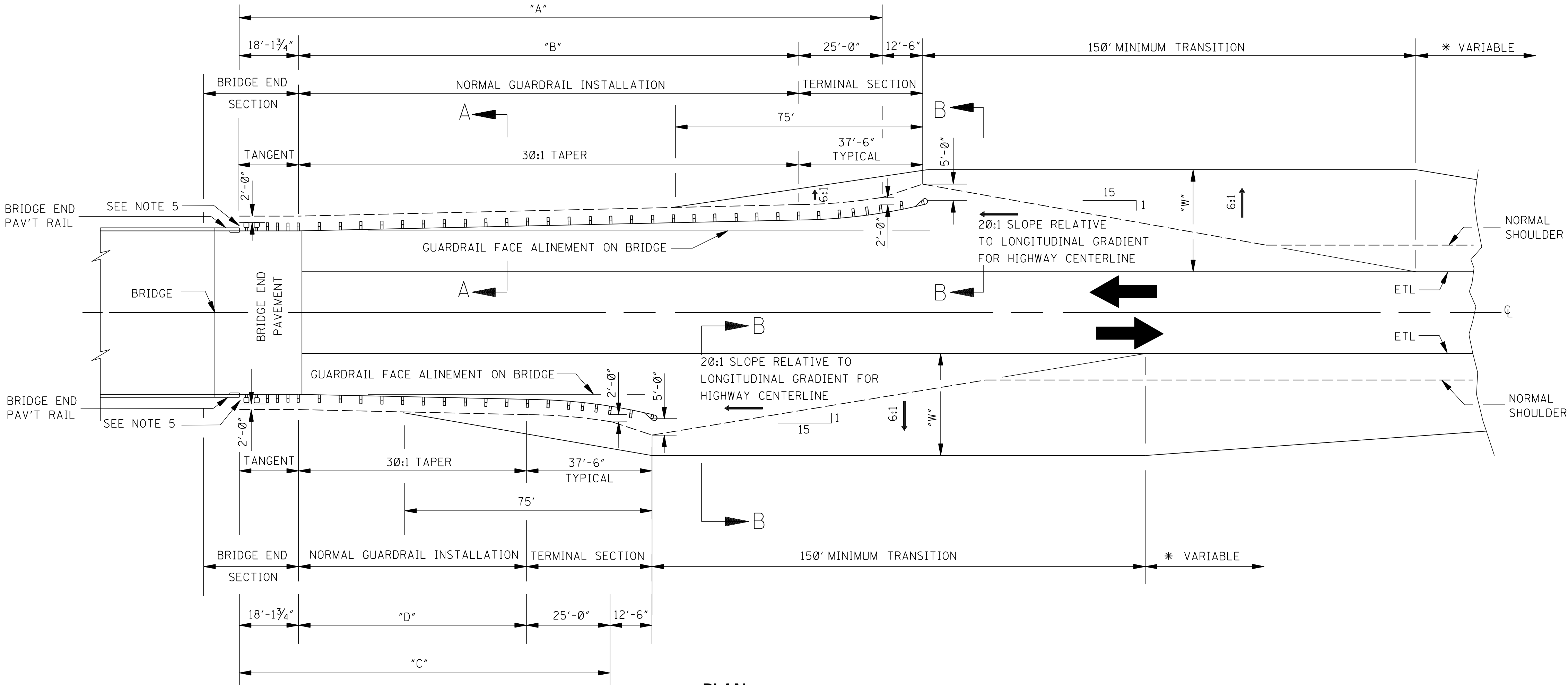
* The distance beyond the hazard should be determined by a length-of-need calculation for opposing traffic, if applicable.

APPENDIX B

MDOT Standard Plan GR-4a - Two-Lane, Two-Way Highways

Embedded reference document included for calculation traceability.

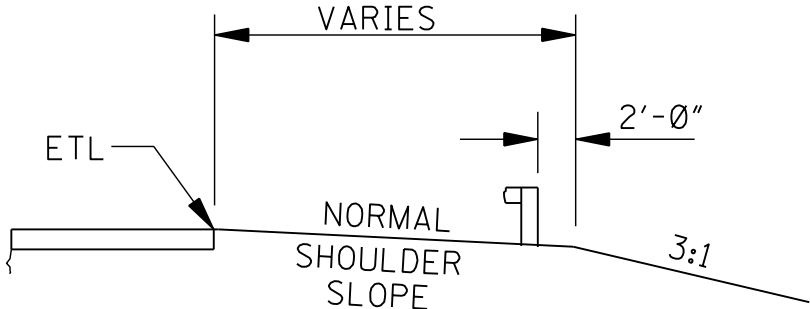
* NOTE: IF FORESLOPE, SHOWN ELSEWHERE ON PLANS, IS OTHER THAN 6:1, TRANSITION WILL OCCUR IN AREA SHOWN.



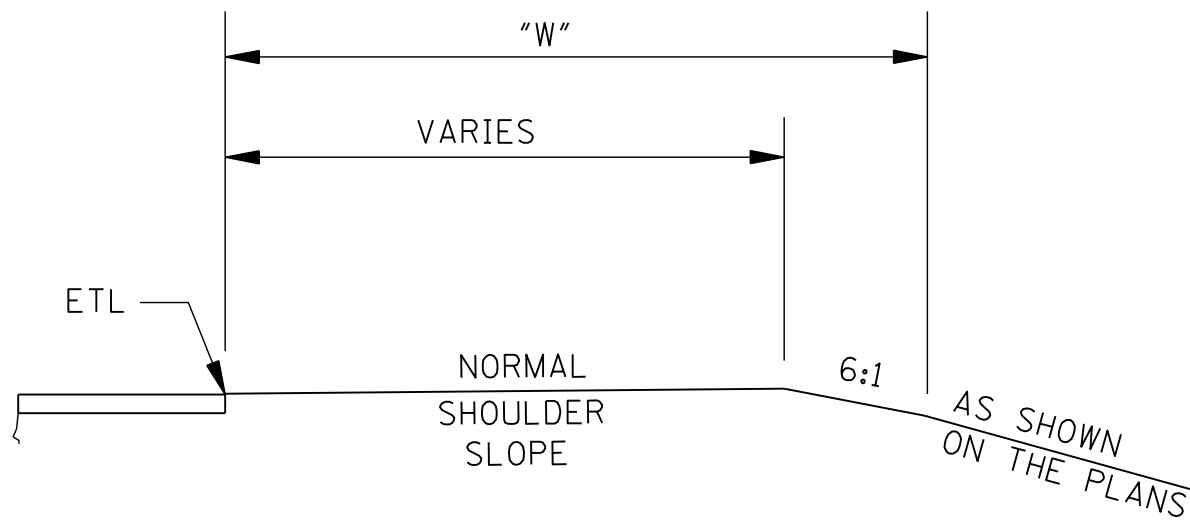
DETAIL OF GUARDRAIL SECTION LAPS

GENERAL NOTES:

- VALUES FOR "A", "B", "C" AND "D" WILL BE SHOWN ELSEWHERE ON THE PLANS.
- FOR DETAILS PERTINENT TO INSTALLATION OF THE TERMINAL SECTION, SEE MANUFACTURER'S SPECIFICATIONS AND DRAWINGS OR ELSEWHERE ON PLANS
- GUARDRAIL SECTIONS SHALL BE LAPPED IN THE DIRECTION OF TRAFFIC APPROACHING THE BRIDGE.
- THE OVERALL LENGTH OF GUARDRAIL IS MEASURED FROM THE CONNECTING END ON THE BRIDGE.
- IN THE ABSENCE OF A BRIDGE END PAVEMENT RAIL, CONNECT THE BRIDGE END SECTION TO THE BRIDGE RAIL (SEE WK. NOS. GR-2 THRU GR-2G). THE SHOULDER WIDTH AT THE BRIDGE END PAVEMENT RAIL OR BRIDGE END RAIL SHOULD BE SUFFICIENTLY WIDE TO PROVIDE A MINIMUM OF 2'-0" BEHIND THE BACK OF POST BEFORE THE SLOPE BREAK (HINGEPOINT).
- TYPE, DETAILS AND LIMITS OF GUARDRAIL BRIDGE END SECTION WILL BE SHOWN ELSEWHERE ON THE PLANS.
- W = SHOULDER WIDTH + FORESLOPE WIDTH. DIMENSIONS FOUND ELSEWHERE ON THE PLANS.



SECTION A-A



SECTION B-B

				BY	MISSISSIPPI DEPARTMENT OF TRANSPORTATION ROADWAY DESIGN DIVISION STANDARD PLAN
				REVISION	GUARDRAIL: TYPICAL INSTALLATION AT BRIDGE APPROACHES FOR 2-LANE, 2-WAY HIGHWAY
				DATE	ISSUE DATE: AUGUST 01, 2017
					 WORKING NUMBER GR-4A SHEET NUMBER 6215